

```

%root_path = fileparts(mfilename("fullpath"));
root_path = pwd;
data_path = fullfile(root_path, 'sym4D_cutsAndFits');

if ~exist('cuts2fit200', 'var')
    ld = load(fullfile(data_path, 'multicuts_fit_dataGH_ei200meV.mat'));
    cuts2fit200 = ld.cuts2fit200;
end
if exist('used_ds200', 'var')
    used_ds = used_ds200;
else
    used_ds = containers.Map()
    sel_info = struct('cuts', [], 'en_range', [], 'fname', '', 'bg_par', []);
    %
    sel_info.cuts = cuts2fit200.cuts_dir001FFfit(1);
    sel_info.en_range = 30:10:110;
    sel_info.cut_dim = 2;
    sel_info.fname='EnFit_Ei200ref001off110_2D_constFFCut';
    sel_info.bg_par= {};
    used_ds('Ei200Br110dir001')=sel_info;
    %
    sel_info.cuts = cuts2fit200.cuts_dir001FFfit(2);
    sel_info.en_range = 30:10:130;
    sel_info.cut_dim = 2;
    sel_info.fname='EnFit_Ei200ref001off200_2D_constFFCut';
    sel_info.bg_par= {};
    used_ds('Ei200Br200dir001')=sel_info;
    %
    sel_info.cuts = {cuts2fit200.cut200_sel};
    sel_info.en_range = 40:10:140;
    sel_info.cut_dim = 2;
    sel_info.fname='EnFit_Ei200Br200dir010_2peaks_2D';
    sel_info.bg_par= {};
    used_ds('Ei200Br200dir010_2peaks')=sel_info;
    %
    sel_info.cuts = {cut(cuts2fit200.cut200_sel, [0,0.02,1], [])};
    sel_info.en_range = 40:10:140;
    sel_info.cut_dim = 2;
    sel_info.fname='EnFit_Ei200Br200dir010_1peak_2D';
    sel_info.bg_par= {};
    used_ds('Ei200Br200dir010_1peak')=sel_info;
    %
    sel_info.cuts = {cut(cuts2fit200.cut110_sel, [0,0.02,2], [])};
    sel_info.en_range = 30:10:110;
    sel_info.cut_dim = 2;
    sel_info.fname='EnFit_Ei200Br110dir010_2D';
    sel_info.bg_par= {};
    used_ds('Ei200Br110dir010')=sel_info;

    %-----
    sel_info.cuts = cuts2fit200.cuts_dir001FFfit(1);
    sel_info.en_range = 30:10:110;

```

```

sel_info.cut_dim = 1;
sel_info.fname='EnFit_Ei200ref001off110_1D_constFFCut';
sel_info.bg_par= {};
used_ds('Ei200Br110dir001_1D')=sel_info;
%
sel_info.cuts = cuts2fit200.cuts_dir001FFfit(2);
sel_info.en_range = 30:10:130;
sel_info.cut_dim = 1;
sel_info.fname='EnFit_Ei200ref001off200_1D_constFFCut';
sel_info.bg_par= {};
used_ds('Ei200Br200dir001_1D')=sel_info;

%-----
sel_info.cuts = cuts2fit200.cut110_selNSfit;
sel_info.en_range = 40:10:120;
sel_info.cut_dim = 2;
sel_info.fname='EnFit_Ei200ref110off100_2D_nonSym';
sel_info.bg_par= {};
used_ds('Ei200Br110dir001_2DNS')=sel_info;

used_ds200 = used_ds;

```

end

used_ds =

Map with properties:

Count: 0
 KeyType: char
 ValueType: any

```

select= 'Ei200Br110dir001_2DNS';
sel_info = used_ds(select);

cuts_list = sel_info.cuts

```

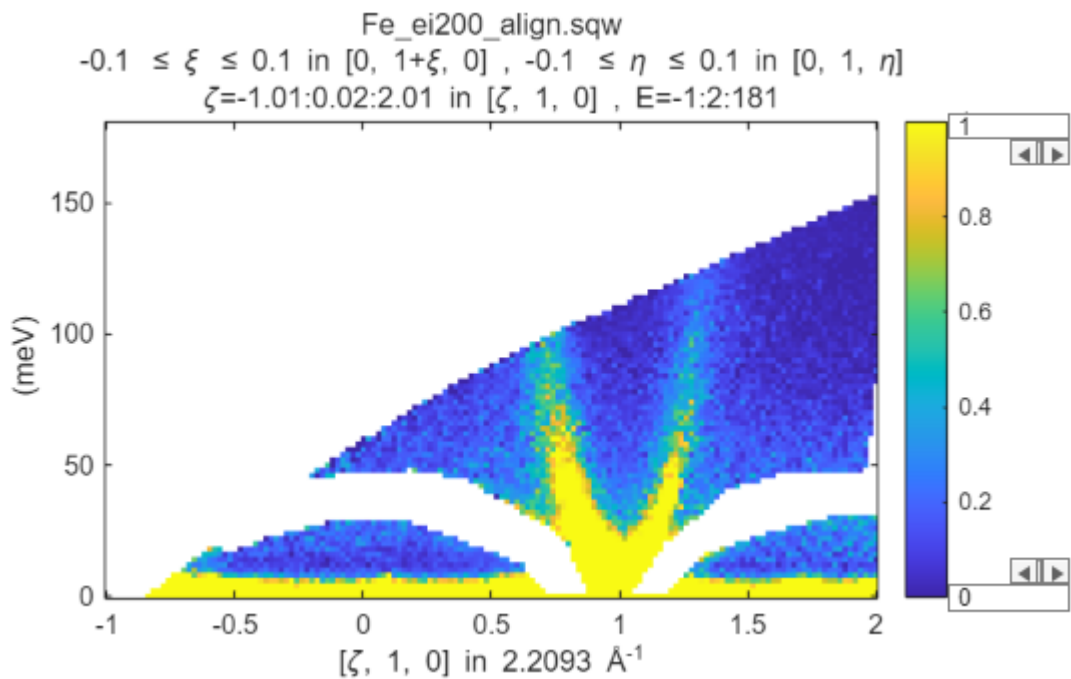
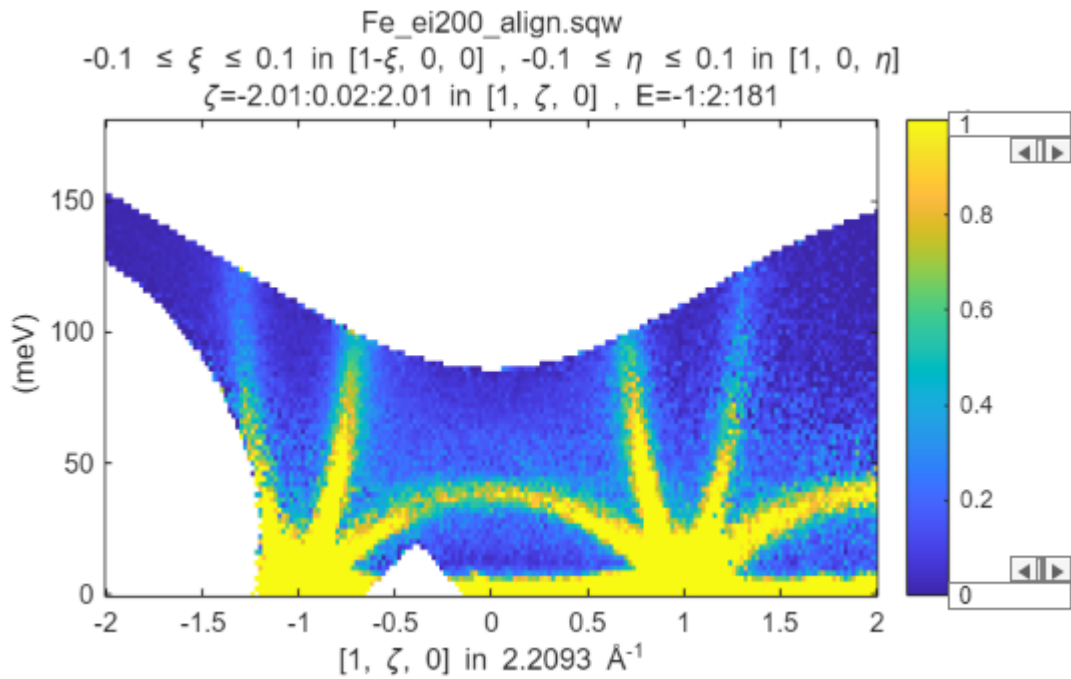
cuts_list = 1x2 cell

	1	2
1	1x1 sqw	1x1 sqw

```

mi = maps_instrument(200,600,'S');
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for i=1:numel(cuts_list)
    cuts_list{i} = cuts_list{i}.set_instrument(mi);
    cuts_list{i} = cuts_list{i}.set_sample(sample);
    plot(cuts_list{i}); lz 0 1; keep_figure;
end

```



```
dir_name = "GH";

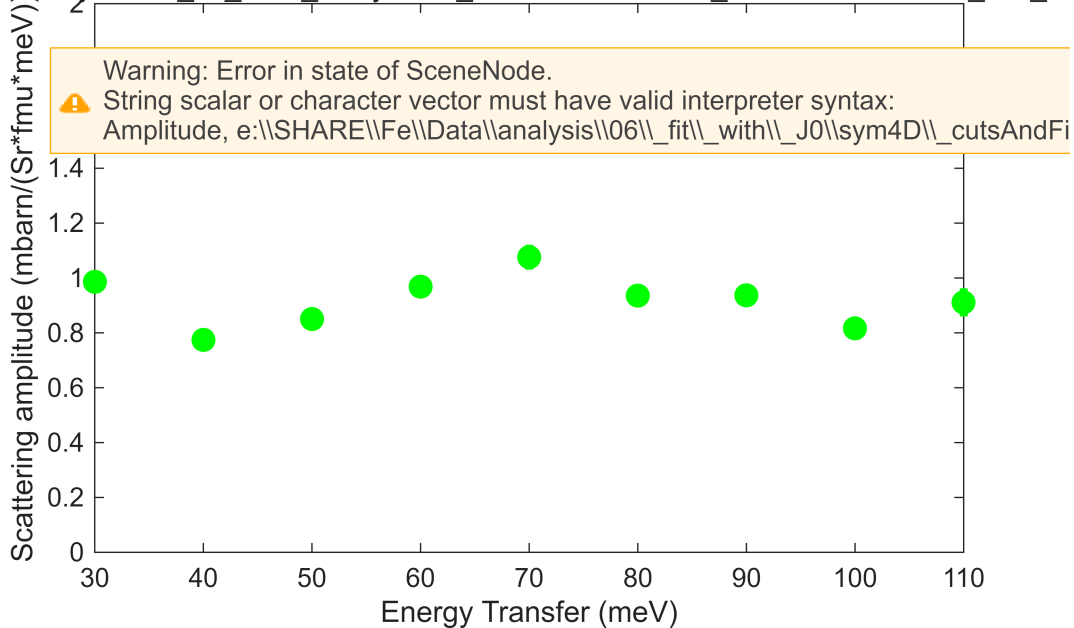
dE_step = 2; %original energy transfer step data were binned to. No point in
going finer
half_dE = 10; % half width of data binning

cut_en = sel_info.en_range;
cut_dimensions = sel_info.cut_dim;

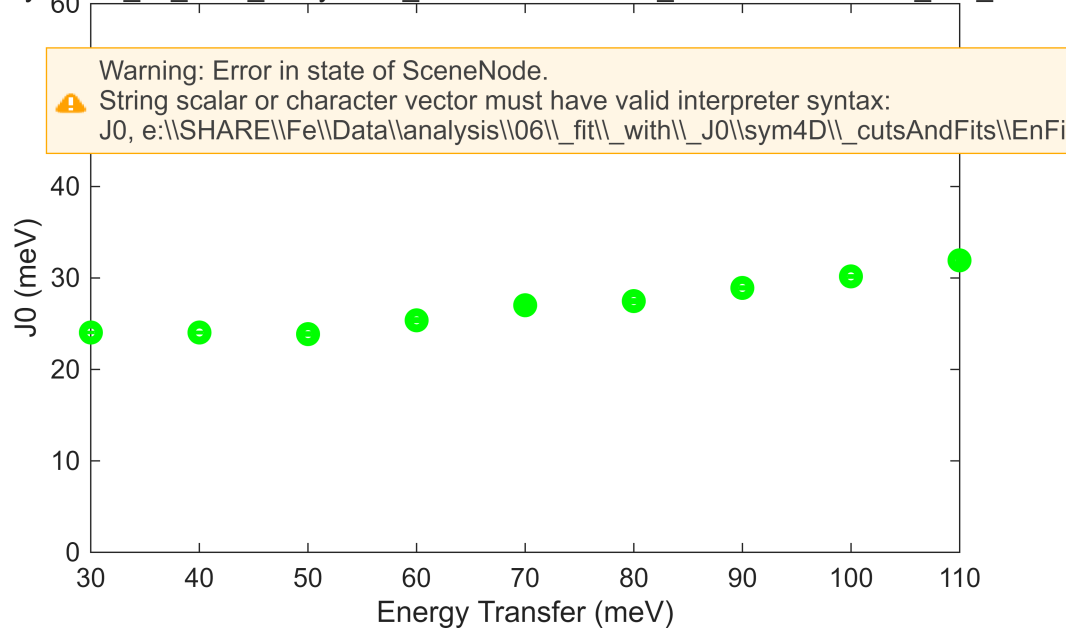
bg_par = sel_info.bg_par;
cuts_list{end+1} = bg_par;
fbase_name = sel_info.fname;
```

```
[fit_res_rec,res_name] = fit_multicuts_along_direction(...
    cuts_list,fbase_name,dir_name,cut_dimensions,cut_en,dE_step,half_dE,true);
```

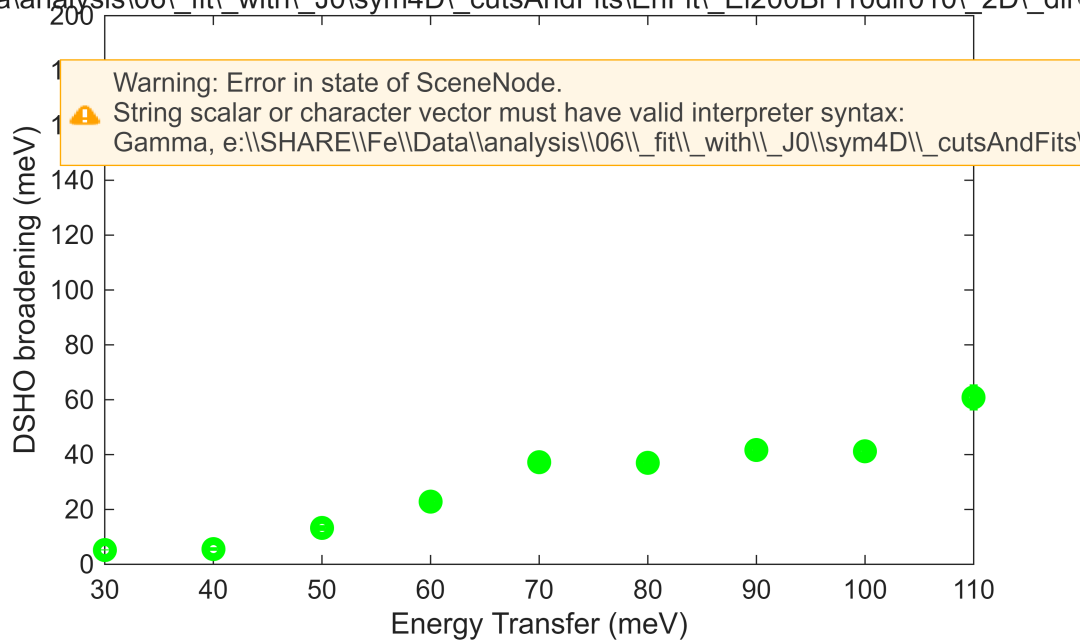
Data\analysis\06\ fit_with_J0\sym4D_cutsAndFits\EnFit_Ei200Br110dir010_2D_di



a\analysis\06\ fit_with_J0\sym4D_cutsAndFits\EnFit_Ei200Br110dir010_2D_dirGH



Data\analysis\06_fit_with_J0\sym4D_cutsAndFits\EnFit_Ei200Br110dir010_2D_dir



**** En = 30 ± 10

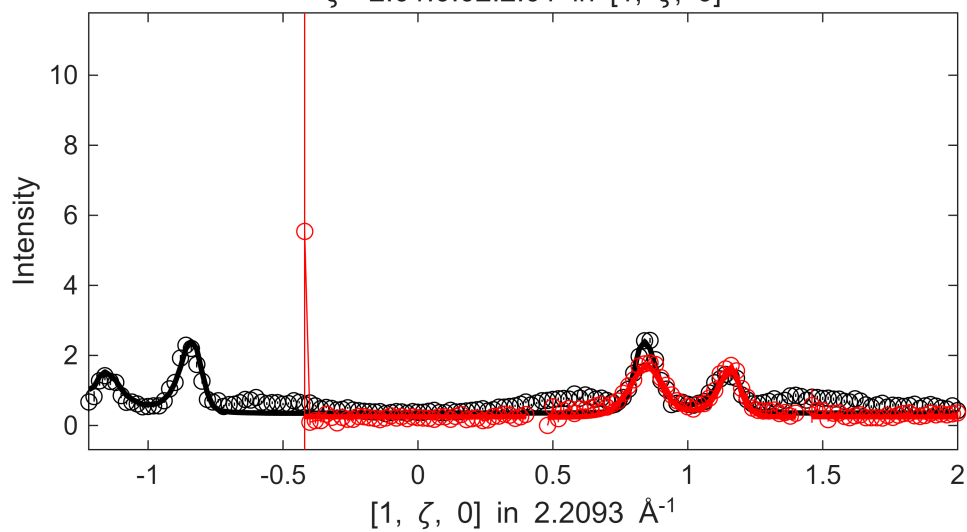
Warning: Dataset 2:3 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.18061

Fe_ei200_align.sqw

S= 0.98 ± 0.021 ; J0= 24 ± 0.21 ; gamma= 4.27669 ± 0.171398

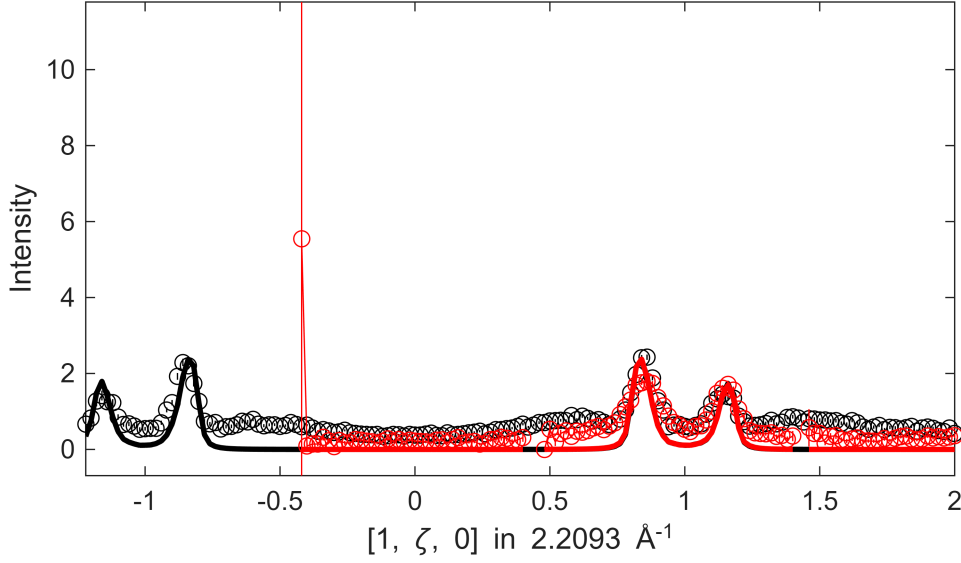
$-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $20 \leq E \leq 40$

$\zeta = -2.01:0.02:2.01$ in $[1, \zeta, 0]$



Fe_ei200_align.sqw

$-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $20 \leq E \leq 40$
 $\zeta = -2.01:0.02:2.01$ in $[1, \zeta, 0]$



**** En = 40 ± 10

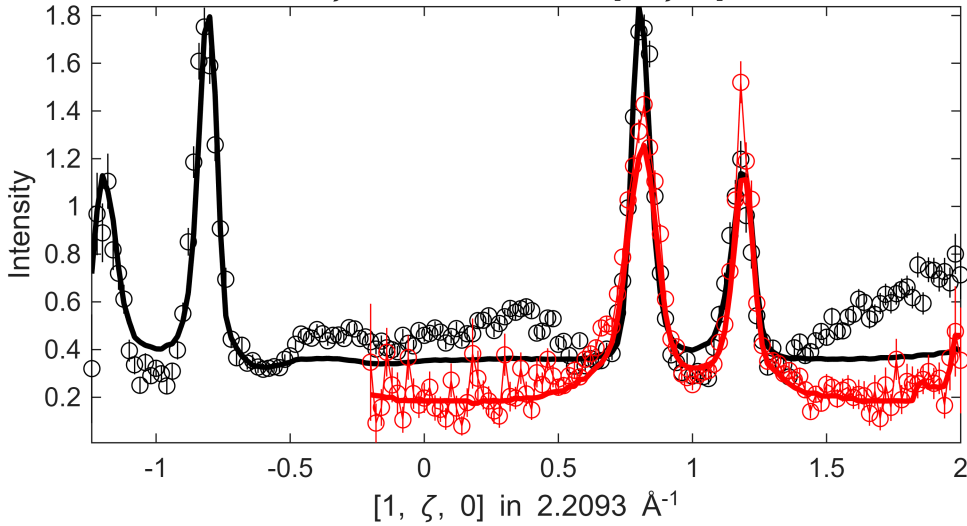
Warning: Dataset 1:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.045228

Warning: Dataset 2:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.12041

Fe_ei200_align.sqw

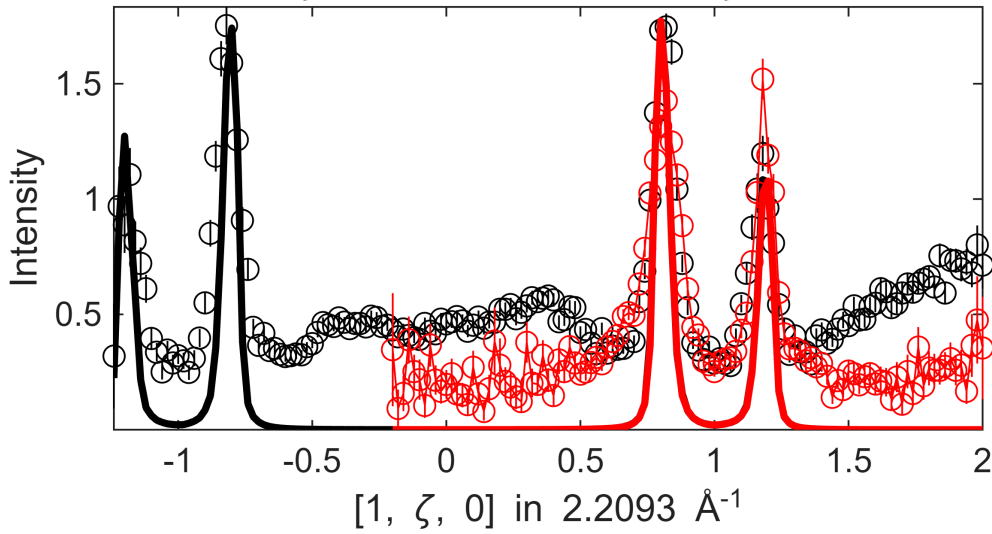
$S = 0.66 \pm 0.018$; $J_0 = 24 \pm 0.16$; $\gamma = 3.51209 \pm 0.379542$

$-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $30 \leq E \leq 50$
 $\zeta = -2.01:0.02:2.01$ in $[1, \zeta, 0]$



Fe_ei200_align.sqw

$\leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $30 \leq \zeta$
 $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$



**** En = 50±10

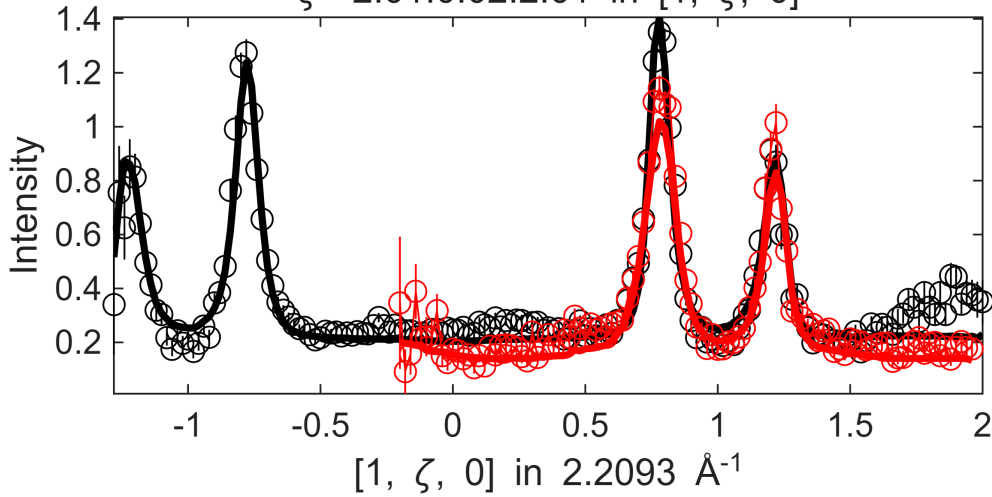
Warning: Dataset 1:8 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.36183

Warning: Dataset 2:8 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.48164

Fe_ei200_align.sqw

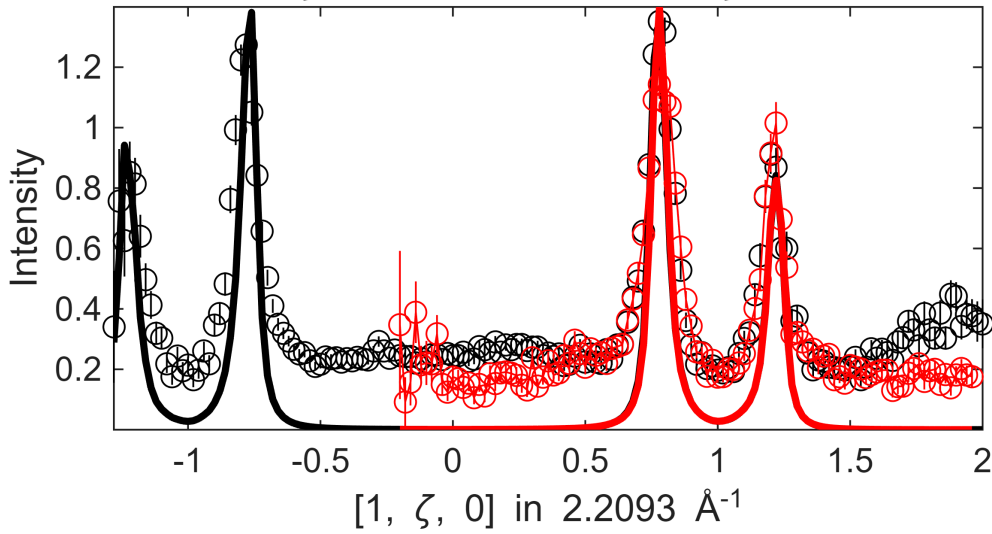
S=0.77±0.015; J0= 24±0.13; gamma=9.72328±0.440025

$\leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $40 \leq \zeta$
 $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$



Fe_ei200_align.sqw

$\leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $40 \leq \zeta$
 $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$



**** En = 60±10

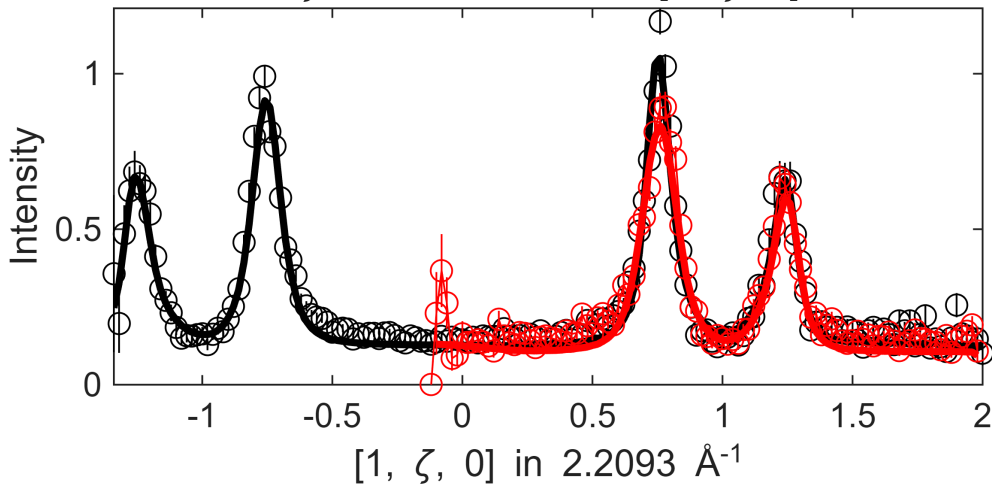
Warning: Dataset 1:36 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.6282

Warning: Dataset 2:11 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.66225

Fe_ei200_align.sqw

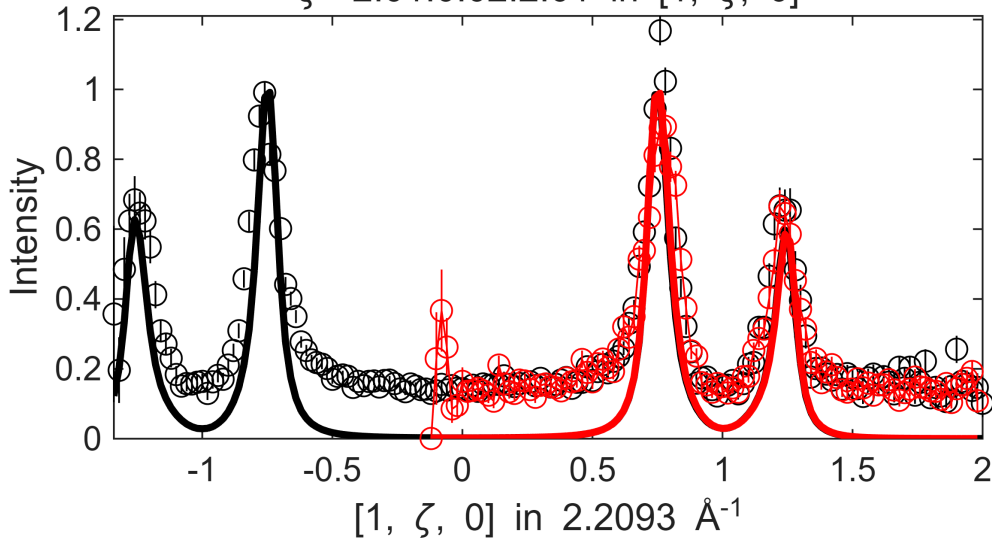
S= 0.9±0.017; J0= 25±0.15; gamma=18.9436±0.672877

$\leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $50 \leq \zeta$
 $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$



Fe_ei200_align.sqw

$\leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $50 \leq \Gamma$
 $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$



**** En = 70±10

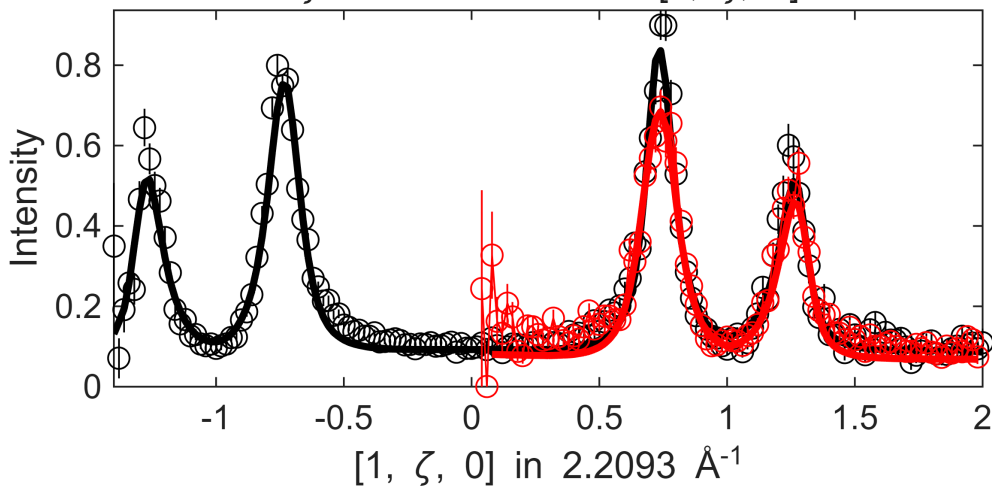
Warning: Dataset 1:61 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 2.7589

Warning: Dataset 2:14 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.84287

Fe_ei200_align.sqw

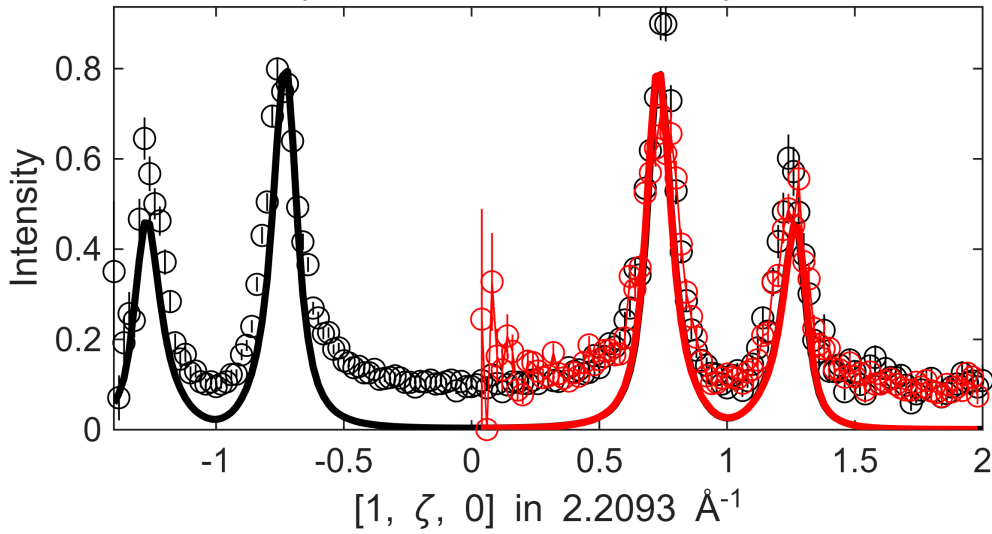
S=0.94±0.014; J0= 26±0.14; gamma=27.1485±0.60331

$\leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $60 \leq \Gamma$
 $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$



Fe_ei200_align.sqw

$\leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $60 \leq \zeta$
 $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$



**** En = 80±10

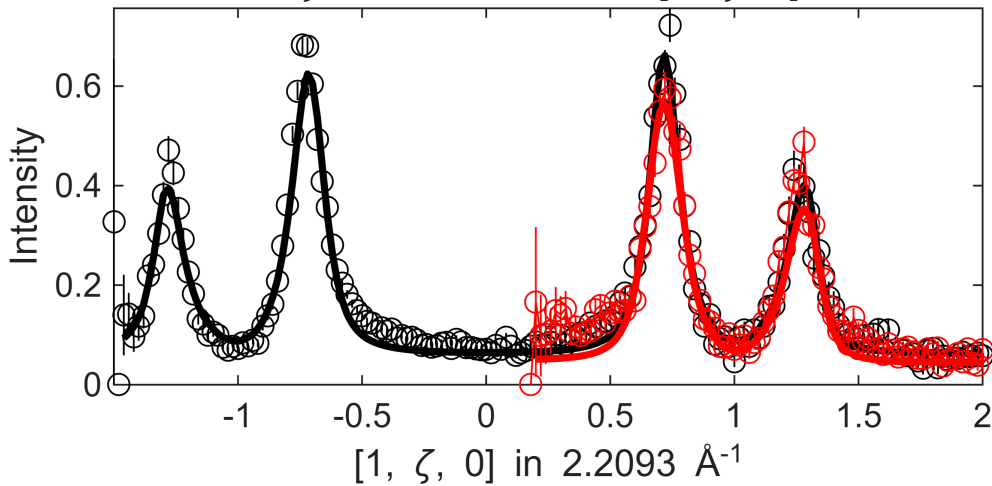
Warning: Dataset 1:115 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 5.2013

Warning: Dataset 2:37 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 2.2276

Fe_ei200_align.sqw

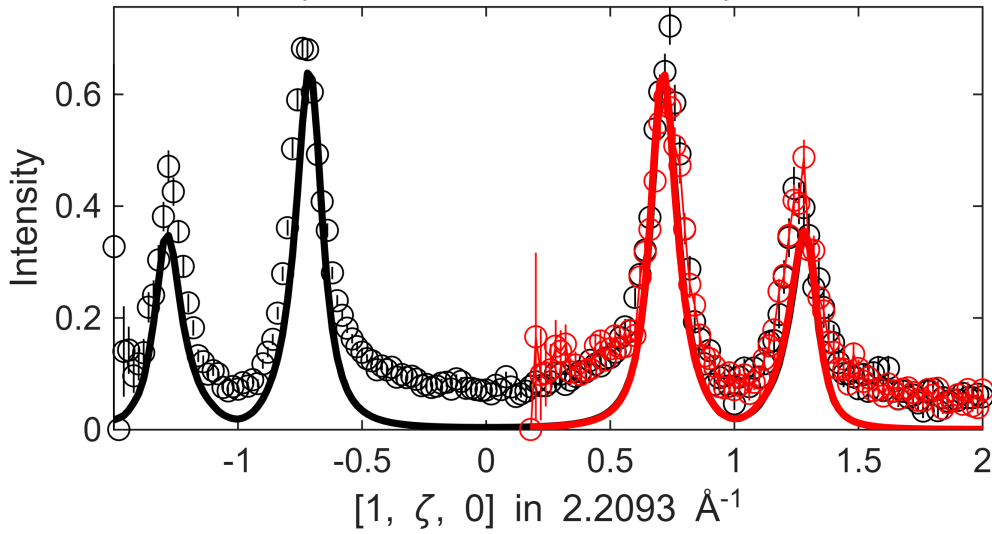
S=0.91±0.019; J0= 28± 0.2; gamma=34.4907±1.17199

$\leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $70 \leq \zeta$
 $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$



Fe_ei200_align.sqw

$\leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $70 \leq E$
 $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$



**** En = 90±10

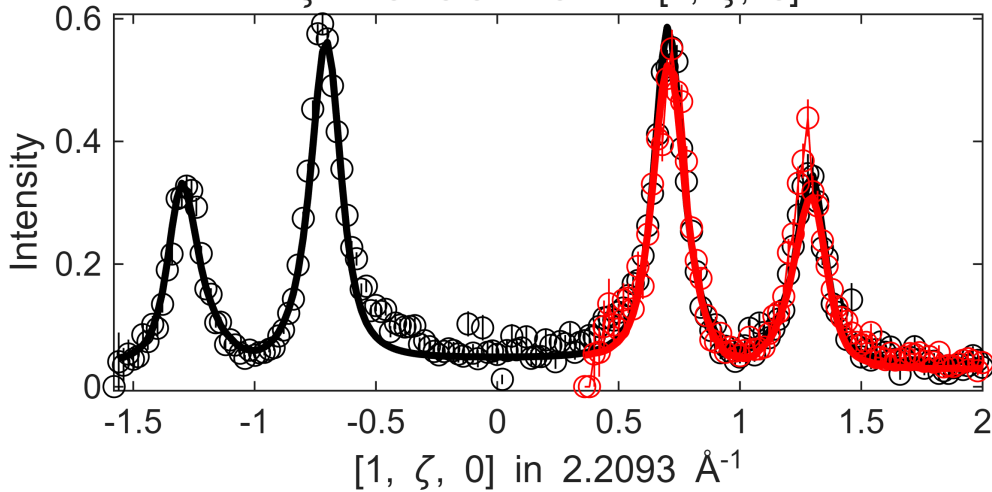
Warning: Dataset 1:150 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 6.7843

Warning: Dataset 2:57 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 3.4317

Fe_ei200_align.sqw

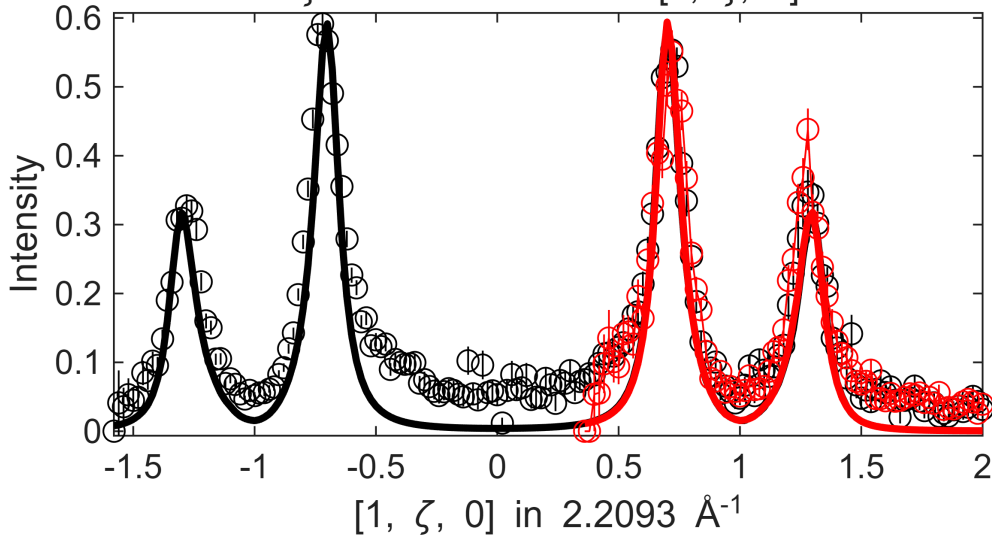
S=0.92±0.017; J0= 28±0.17; gamma=37.8656±1.12215

$\leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $80 \leq E$
 $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$



Fe_ei200_align.sqw

$\leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $80 \leq E$
 $\zeta = -2.01:0.02:2.01$ in $[1, \zeta, 0]$



**** En = 100±10

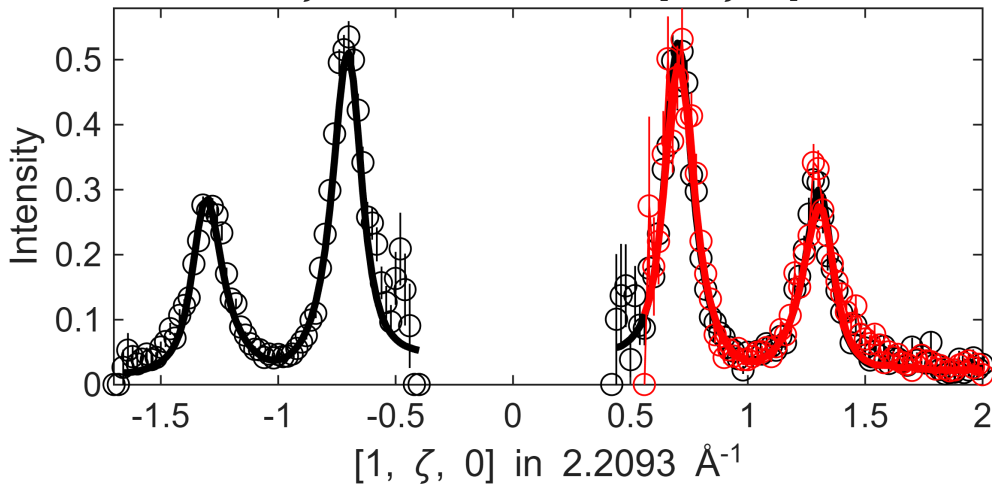
Warning: Dataset 1:148 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 6.6938

Warning: Dataset 2:66 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 3.9735

Fe_ei200_align.sqw

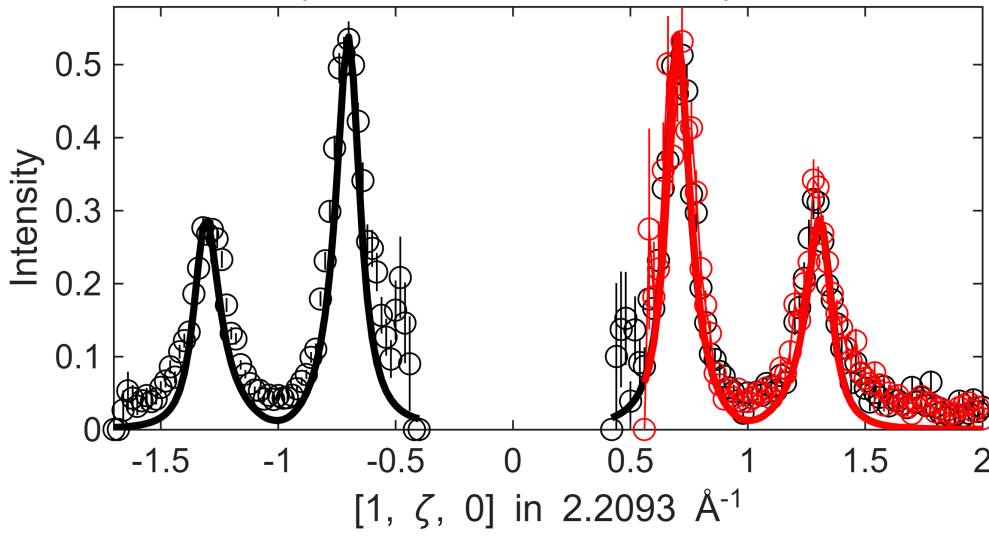
S=0.89±0.019; J0= 30±0.18; gamma=39.4735±1.28552

$\leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $90 \leq E$
 $\zeta = -2.01:0.02:2.01$ in $[1, \zeta, 0]$



Fe_ei200_align.sqw

$\leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $90 \leq E$
 $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$



**** En = 110±10

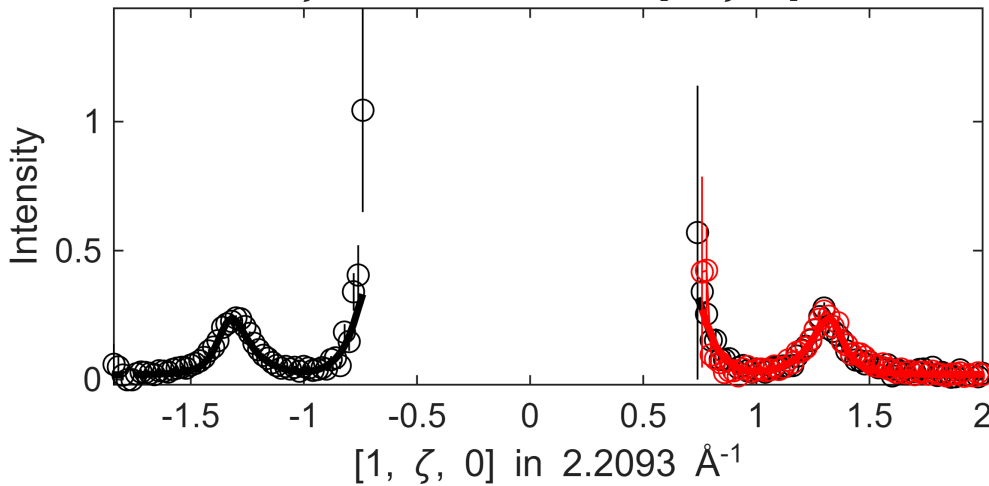
Warning: Dataset 1:155 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 7.0104

Warning: Dataset 2:85 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 5.1174

Fe_ei200_align.sqw

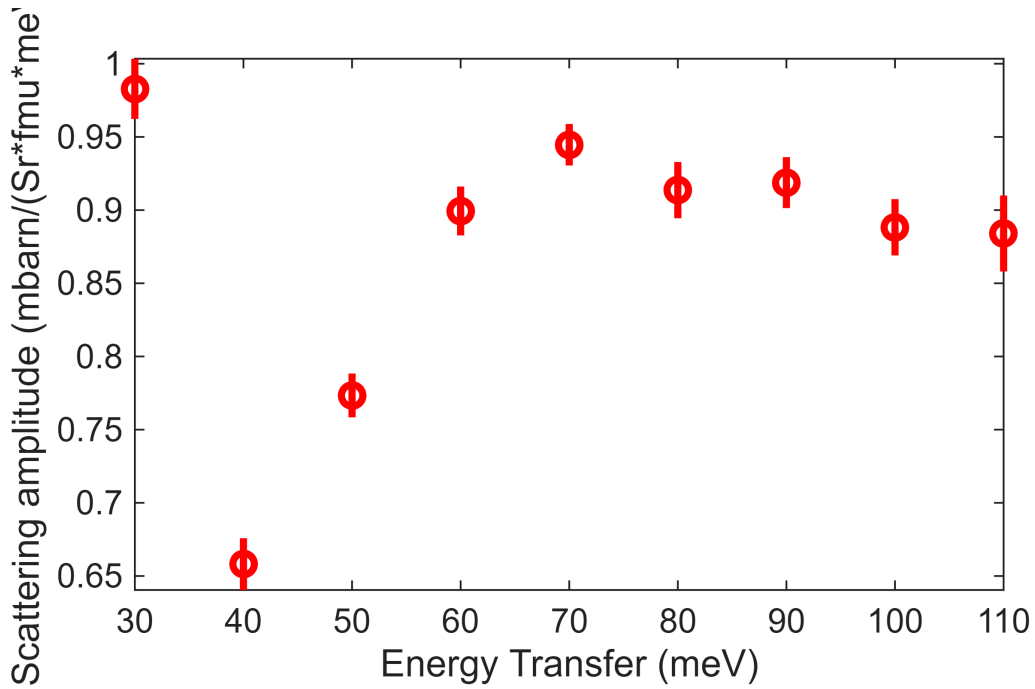
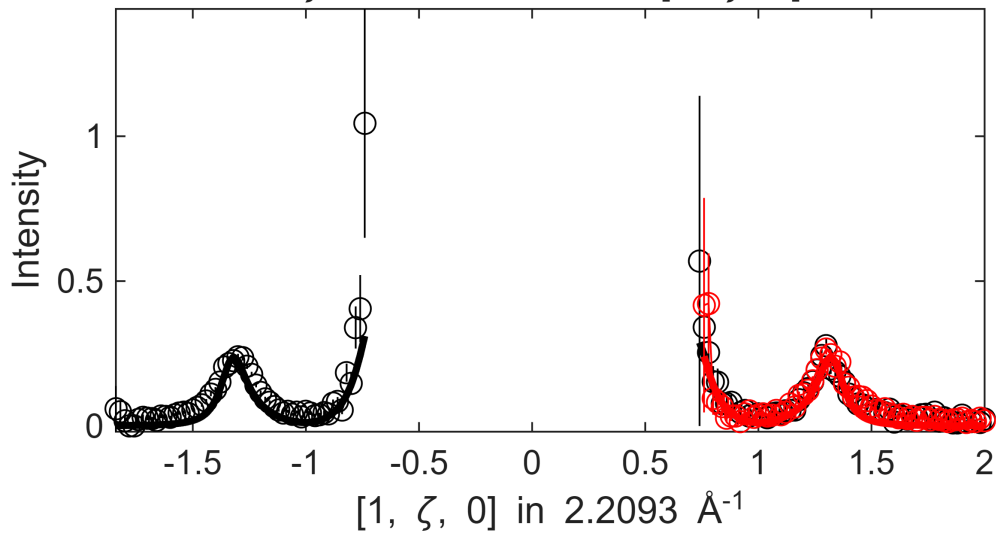
S=0.88±0.026; J0= 31±0.25; gamma=47.7192±2.0805

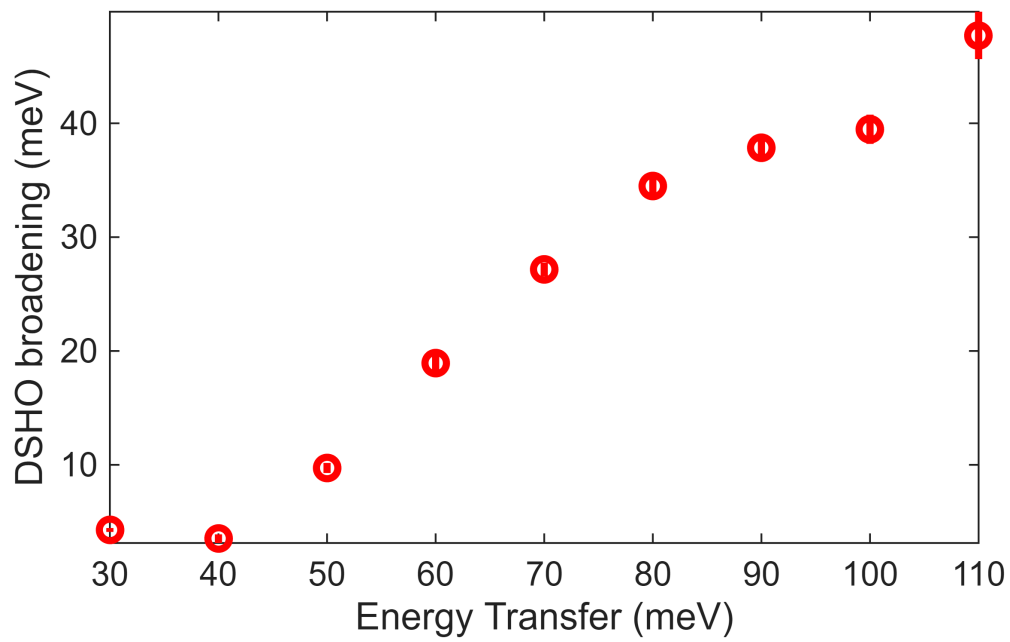
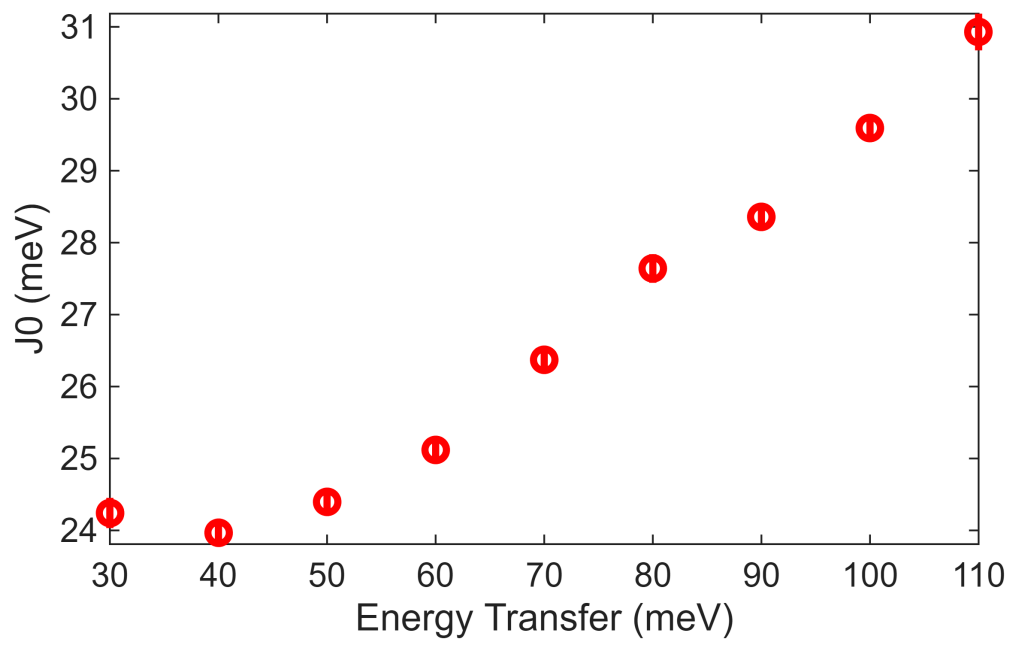
$\leq \xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $100 \leq E$
 $\zeta=-2.01:0.02:2.01$ in $[1, \zeta, 0]$

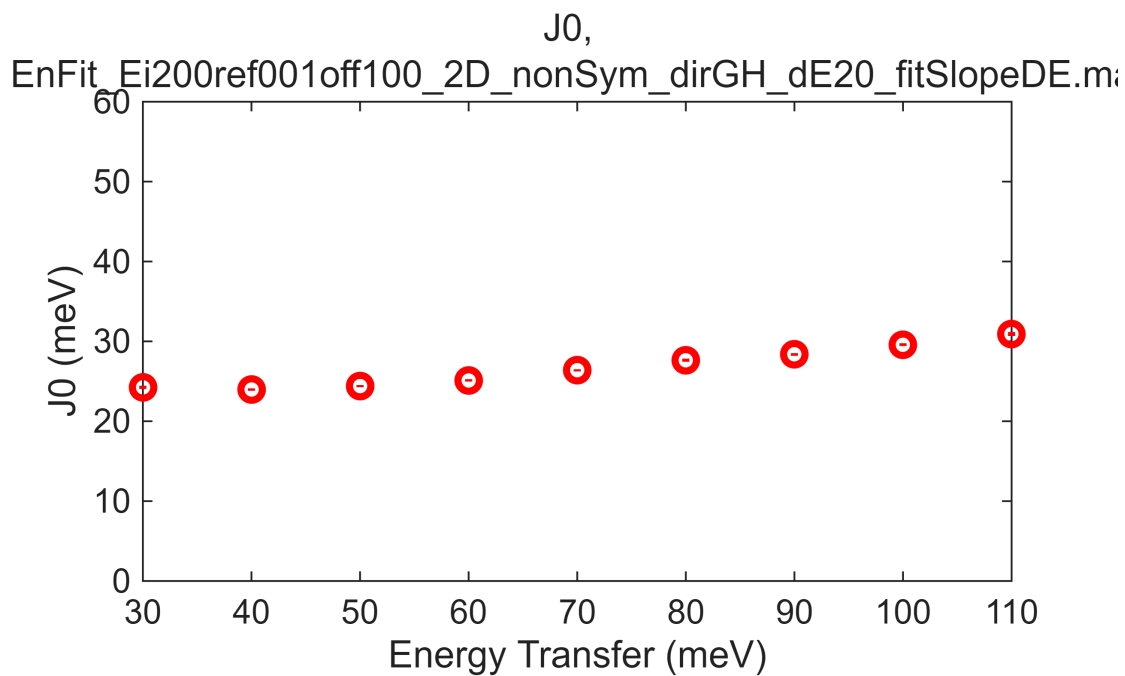
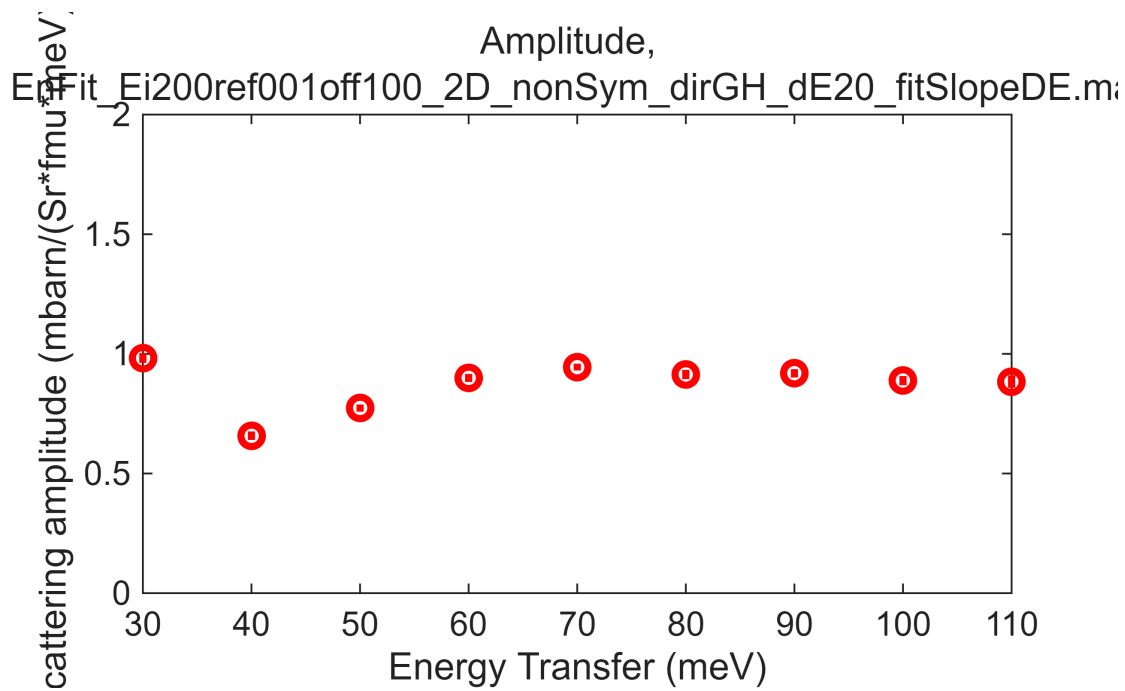


Fe_ei200_align.sqw

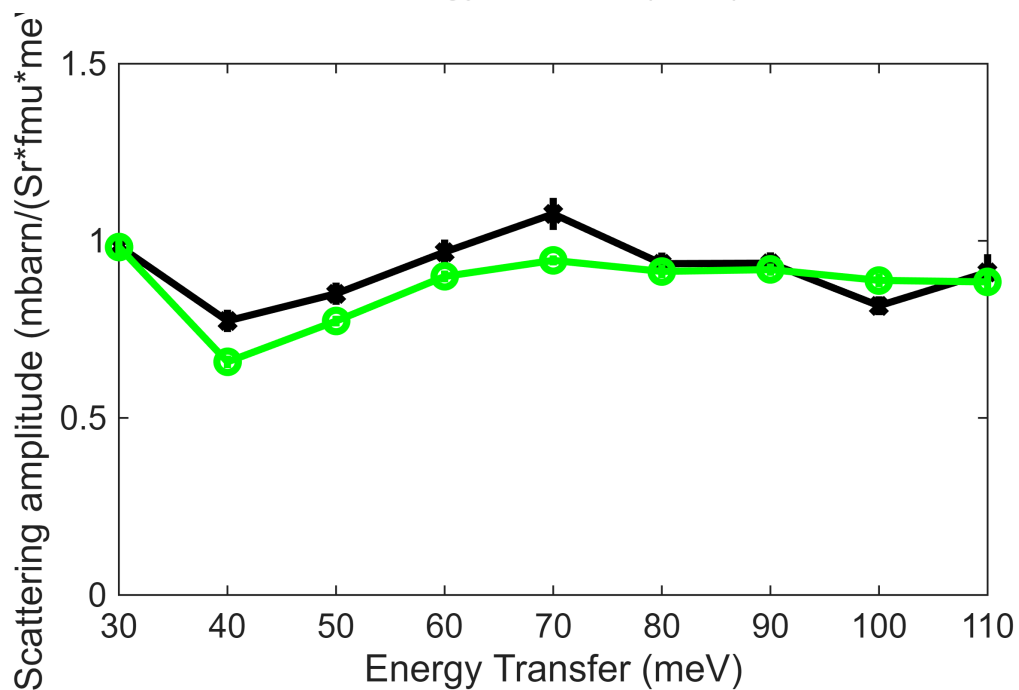
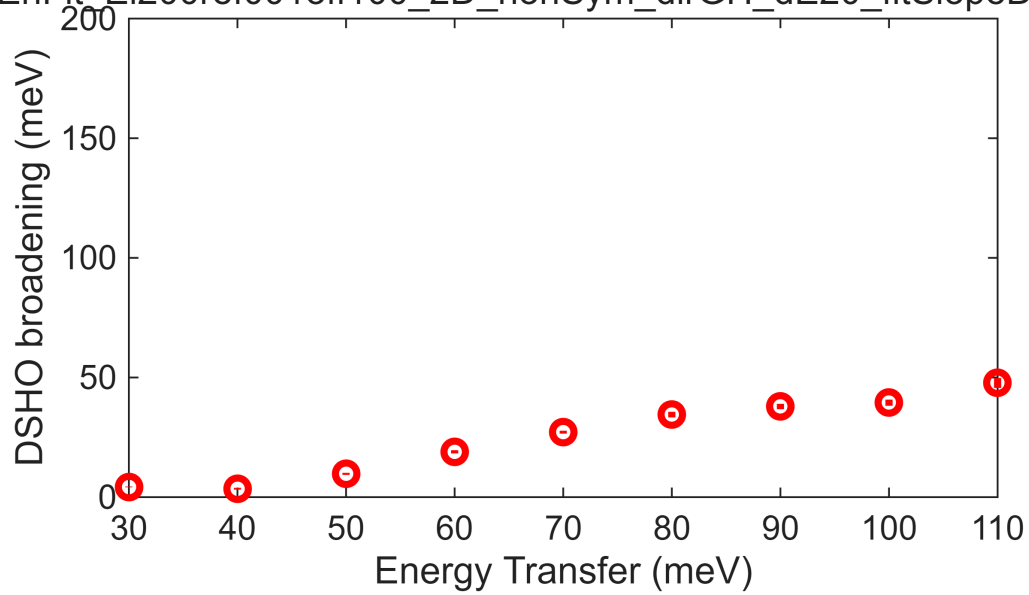
$\xi \leq 0.1$ in $[1-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 0, \eta]$, $100 \leq \Gamma$
 $\zeta = -2.01:0.02:2.01$ in $[1, \zeta, 0]$

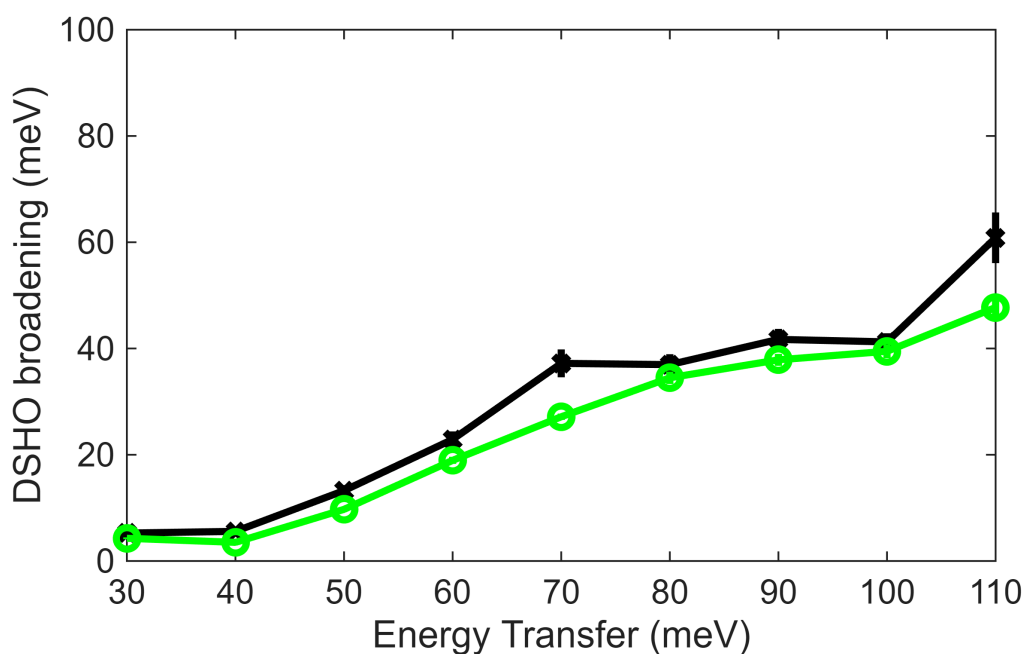
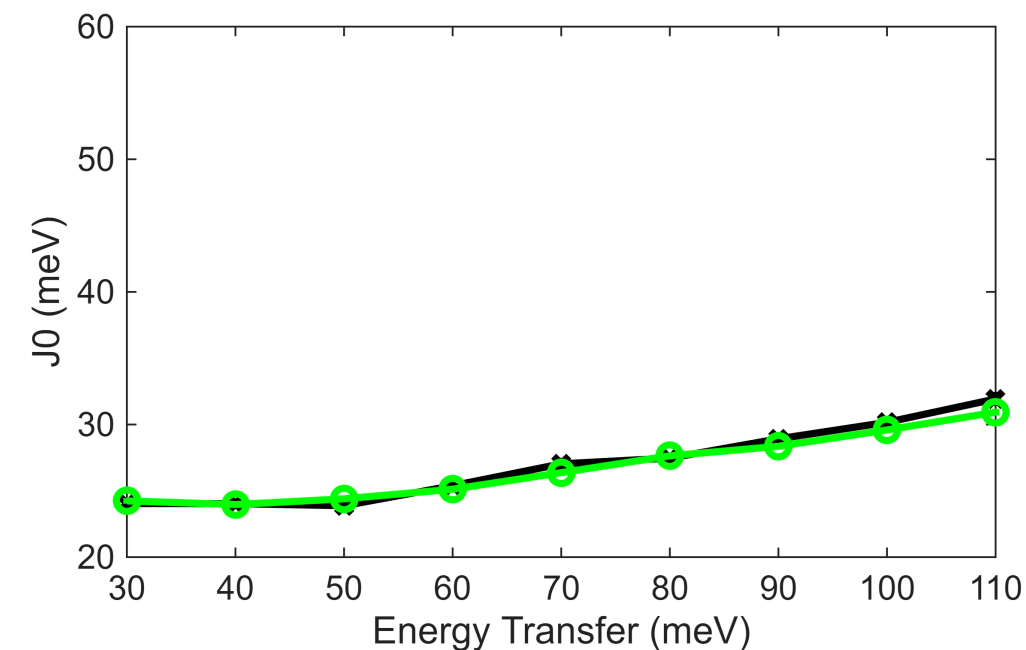






Gamma,
EnFit_Ei200ref001off100_2D_nonSym_dirGH_dE20_fitSlopeDE.m





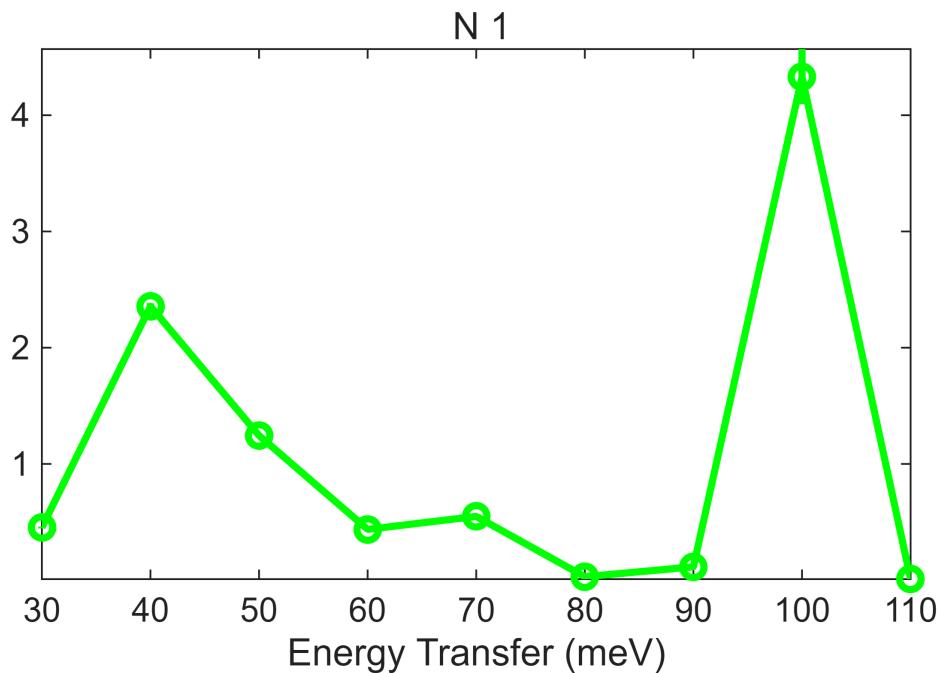
Warning: Error in state of SceneNode.
String scalar or character vector must have valid interpreter syntax:
Amplitude,
e:\SHARE\Fe\Data\analysis\06_fit_with_J0\sym4D_cutsAndFits\EnFit\Ei200Br110dir010_2D_dirGH_dE10_fitSlopeDE.mat
Warning: Error in state of SceneNode.
String scalar or character vector must have valid interpreter syntax:
J0,
e:\SHARE\Fe\Data\analysis\06_fit_with_J0\sym4D_cutsAndFits\EnFit\Ei200Br110dir010_2D_dirGH_dE10_fitSlopeDE.mat
Warning: Error in state of SceneNode.
String scalar or character vector must have valid interpreter syntax:
Gamma,
e:\SHARE\Fe\Data\analysis\06_fit_with_J0\sym4D_cutsAndFits\EnFit\Ei200Br110dir010_2D_dirGH_dE10_fitSlopeDE.mat

beep
beep
beep

```

bg_data = extract_bg_par(fit_res_rec.all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(fit_res_rec.all_fit_par(1).en,fit_res_rec.all_fit_par(end).en);
keep_figure;

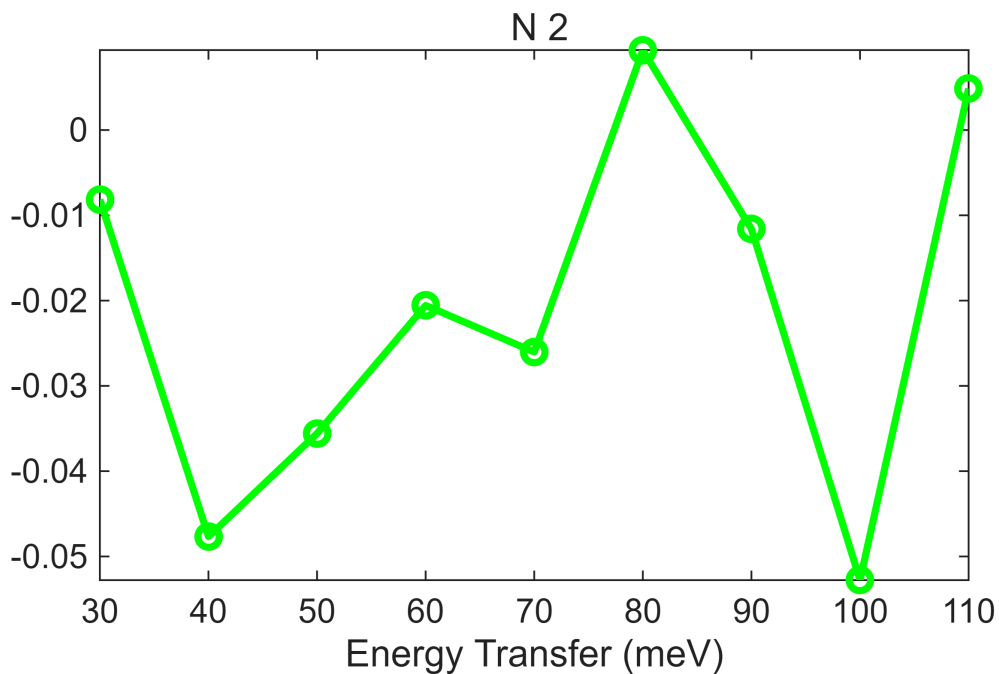
```



```

pd(bg_data(2));lx(fit_res_rec.all_fit_par(1).en,fit_res_rec.all_fit_par(end).en);
keep_figure

```



```

%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_
figure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_
figure

```

```
fit_resEi200 = fit_res_rec;  
save(res_name, "fit_resEi200")
```